

Global Valuation

Cross-Country Valuation Multiples

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Outline

Data and Sample

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Finding 2: P/E Is Cleaner Than M/B

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Data and Sample

Data Sources

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (122 countries)
compustat_america	0.8 GB	254K	984	North America fundamentals
returns_global	1.9 GB	10.4M	25	Global monthly security prices

- **Combined panel:** 743,087 firm-years, 72,720 firms, 81 countries (1986–2024)
- All financial data in *local currency* (millions)
- **Market cap (Global):** $cshoi \times prccm$ using $prirow$ (primary security)

Variable Definitions

Dependent variables:

$$\log(P/E)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Net Income}_{i,t}}\right), \quad NI > 0$$

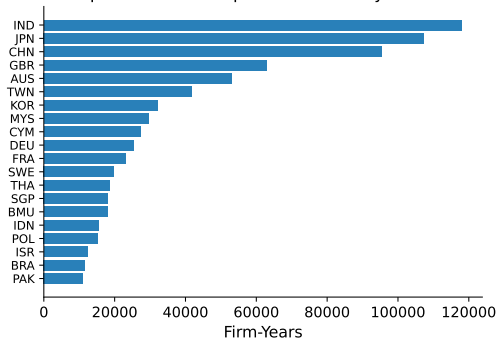
$$\log(M/B)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Book Equity}_{i,t}}\right)$$

Market Cap	US: mkvalt. Global: cshoi $\times$ prccm (primary security)
Book Equity	ceq (common equity). <i>Does not include debt.</i>
Net Income	ib (income before extraordinary items)
Leverage	(at – ceq)/at = non-equity share of assets
ROA	ib / at. Industry: GICS Industry Group (~25 categories)

All ratios winsorized at 1st/99th percentiles. All data in local currency (millions).

Coverage and Descriptive Statistics

Compustat Global: Top 20 Countries by Firm-Years



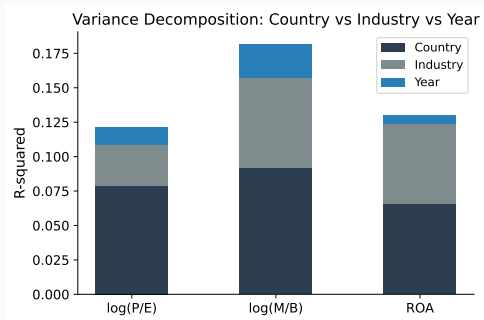
Variable	N	Median	P10	P90
log(P/E)	532K	2.79	1.76	4.32
log(M/B)	743K	0.38	-0.78	1.73
ROA	743K	0.02	-0.15	0.11
Leverage	743K	0.49	0.14	0.85

Top 5 by obs: USA (188K), JPN (60K), CHN (58K), IND (57K), TWN (31K).

P/E restricted to profitable firms (72%). Global prices start 2007.

Finding 1: Country Dominates Valuation

Variance Decomposition: Country vs Industry vs Year



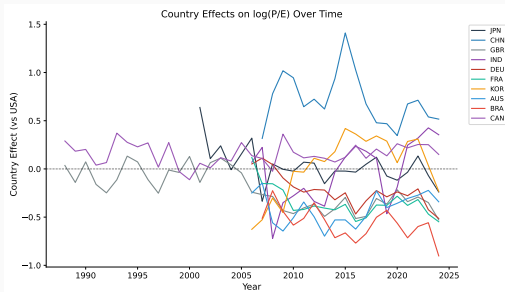
Shapley R^2 decomposition:

	P/E	M/B	ROA
Country	64.6%	50.8%	50.7%
Industry	24.8%	35.8%	44.8%
Year	10.6%	13.4%	4.5%

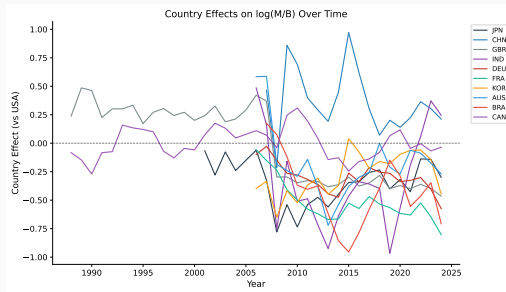
Country dominates all three. P/E shows the *strongest* country dependence.

Extends Heston–Rouwenhorst (1994) from *returns* to *valuation levels*.

Country Effects on P/E and M/B (vs USA)



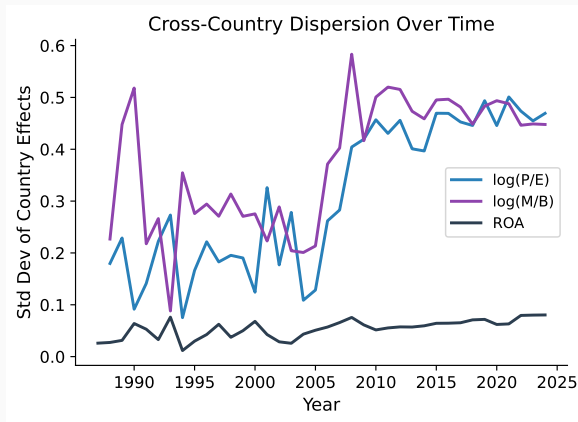
P/E country effects



M/B country effects

Year-by-year regressions with size, leverage, earnings growth, industry FE. USA = 0. China premium is largest but declining; Japan stable near zero (P/E) or deep discount (M/B).

Valuation Dispersion Is Diverging



Dispersion is diverging:

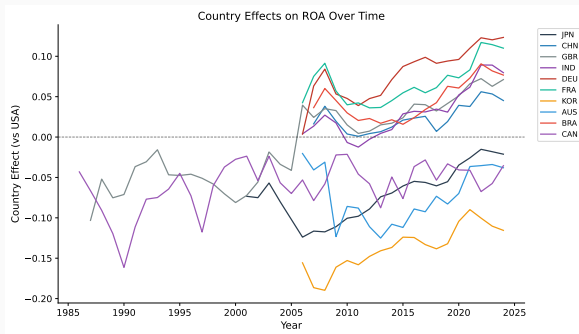
- P/E: 0.17 $\rightarrow$ 0.47 ($\times 2.7$)
- M/B: 0.40 $\rightarrow$ 0.45

ROA dispersion more stable
(~ 0.05 – 0.10), with a spike during
the GFC.

Despite financial globalization, country
valuations are *not converging*.

Finding 2: P/E Is Cleaner Than M/B

Japan: The P/E vs M/B Puzzle



Japan's “discount” depends on the metric:

- M/B: -0.35 (negative 100% of years)
- P/E: $+0.03$ (near zero)
- ROA: positive (high profitability)

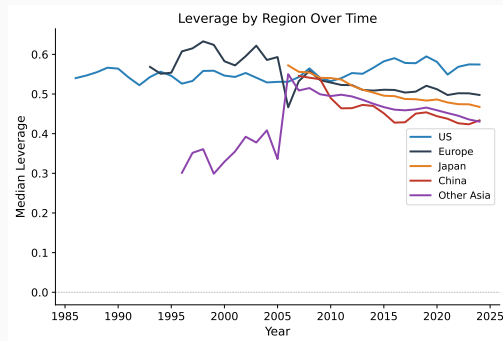
Japan's M/B discount reflects *conservative book values* (domestic GAAP, 93%), not lower market valuations vs earnings.

P/E is a cleaner cross-country comparison.

M/B Is Contaminated by Leverage

Finding	P/E	M/B
Country R^2 share	64.6%	50.8%
Leverage coeff	-0.02	+0.90***
Japan effect	+0.03	-0.35
Dispersion trend	$\times 2.7$	$\times 1.1$

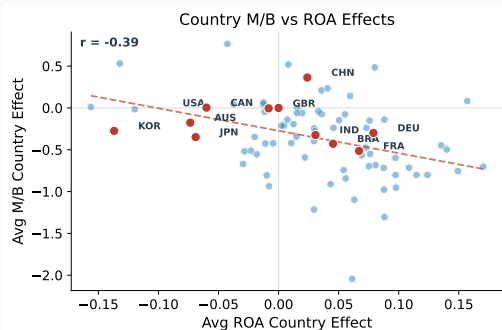
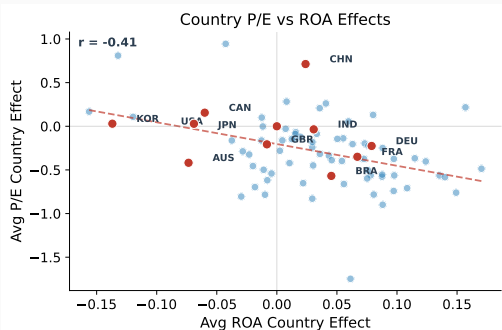
Leverage is +0.90*** in M/B (mechanical: less equity $\Rightarrow$ higher M/B) but insignificant in P/E. Cross-country leverage std = 0.095, mechanically inflating M/B variation.



Regional leverage over time

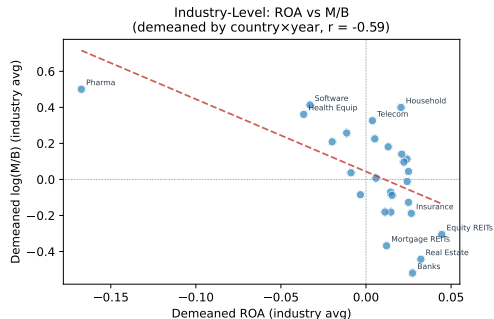
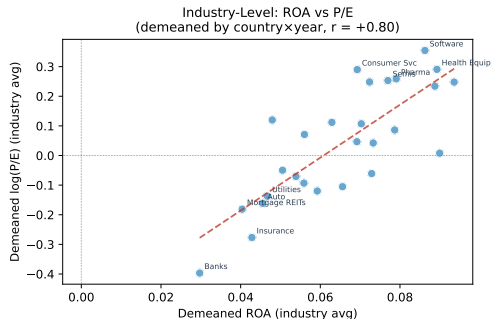
Finding 3: A Profitability–Valuation Puzzle

High-Valuation Countries Are Not More Profitable



Each dot = a country (average effect vs USA). The negative correlation ($r = -0.41$ for P/E) means high-valuation countries have *lower* profitability. Brazil: discounted yet profitable. China: high P/E, average ROA.

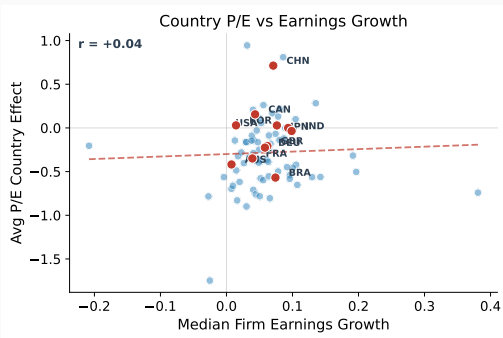
But Within Countries, High Profitability Raises Valuations



Each dot = a GICS industry group (27 industries, demeaned by country $\times$ year). **P/E** ($r = +0.80$): within a country, more profitable industries have higher P/E. **M/B** ($r = -0.59$): reverses (Pharma: low ROA, high M/B from intangibles).

The sign flip (within-country positive, between-country negative) deepens the puzzle.

Earnings Growth Does Not Explain Valuations Either



Cross-country correlations:

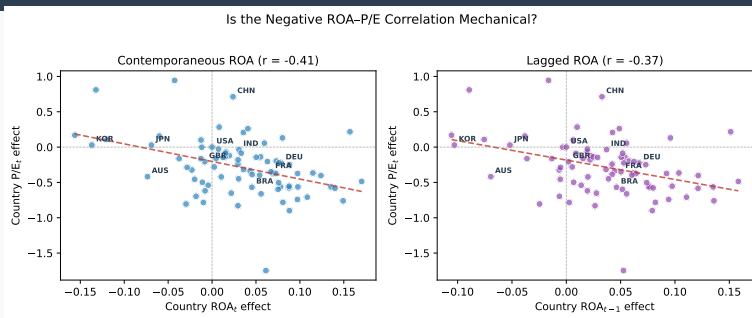
	P/E	M/B
vs ROA	-0.41	-0.38
vs Earn. Growth	-0.09	+0.03

Earnings growth is *uncorrelated* with valuations. Japan has higher growth than the US yet near-zero P/E premium.

Implication: the cross-country valuation spread is a *discount rate* phenomenon, not a cash flow story.

Robustness of the Puzzle

The Correlation Is Not Mechanical (Lagged ROA Test)



Concern: NI is in both ROA (numerator) and P/E (denominator) — a temporary earnings shock mechanically raises ROA and lowers P/E.

Test: replace ROA_t with ROA_{t-1} . Last year's NI is not in this year's P/E.

Result: correlation barely changes ($-0.41 \rightarrow -0.37$). Country ROA effects are extremely persistent ($r = 0.99$), so the link reflects *permanent* profitability differences.

Not Industry Composition Either

Specification	ROA vs P/E	Note
Raw country averages	-0.02	No controls
Industry×year demeaned	+0.11	Industry removed, no other controls
Regression without industry FE	-0.46	Size + leverage + earn. growth
Regression with industry FE	-0.41	Our main specification
Using lagged ROA_{t-1}	-0.37	Breaks NI→P/E mechanical link

- Industry FE slightly *reduce* the correlation ($-0.46 \rightarrow -0.41$) — industry composition is not the driver
- The negative link emerges after controlling for firm size (see next slide)
- Raw correlation is near zero because size heterogeneity masks the relationship

It Is a Simpson's Paradox: Size Controls Reveal the Correlation

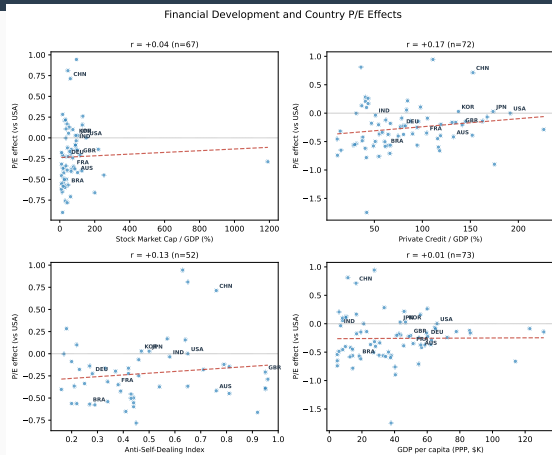
P/E controls	ROA controls	$r(\text{ROA}, \text{P/E})$	
		Local	USD
None	None	−0.02	−0.02
+ size	None	—	−0.14
None	+ size	—	−0.15
+ size	+ size	− 0.51	− 0.64
+ size + leverage	+ size + leverage	—	−0.65
Full spec	+ size + industry	−0.47	−0.59

The negative ROA–P/E correlation is **zero** without size controls. It appears *only* when both P/E and ROA regressions control for firm size. The effect is **stronger** with USD-converted size ($r = -0.64$) — not a currency artifact.

Interpretation (Simpson's paradox): within each size bucket, high-ROA countries have lower P/E. But across size buckets, the relationship is masked.

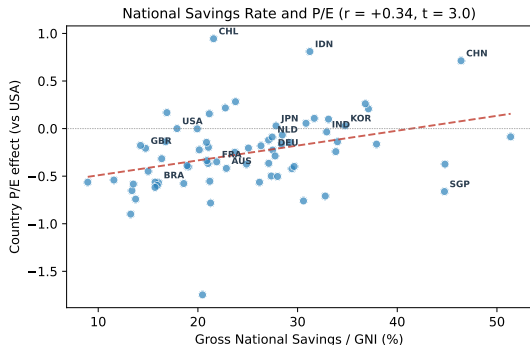
What Explains Cross-Country Valuations?

Standard Institutional Variables Do Not Predict P/E



Market cap/GDP ($r = +0.04$), credit/GDP ($r = +0.17$), anti-self-dealing ($r = +0.13$), GDP per capita ($r = +0.01$): all near zero. La Porta et al. (2002) variables do not explain valuation levels.

National Savings Rate Predicts Valuations



Gross savings / GNI (World Bank):

	P/E	M/B
Savings/GNI	+0.34*** ($t = 3.01$)	+0.29** ($t = 1.89$)

vs ROA -0.12

Robust: excl. Gulf states $r = +0.41$ ($t = 3.40$). Survives controlling for ROA ($t = 2.85$) and GDP per capita ($t = 2.43$).

Interpretation: countries that save more have more capital seeking equity returns, pushing up price multiples. A *demand-side* channel.

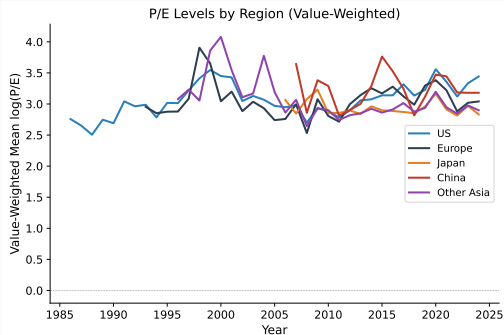
Summary: What Predicts Country P/E Effects?

Variable	Corr with		Partial r (ROA, P/E var)	N
	P/E	ROA		
<i>None (raw)</i>	—	—	−0.41	81
Savings / GNI	+0.34***	−0.12	−0.49	70
Stock Market Cap / GDP	+0.04	−0.17	−0.50	67
Private Credit / GDP	+0.17	−0.20	−0.45	72
Anti-Self-Dealing Index	+0.13	−0.24	−0.55	52
GDP per capita (PPP)	+0.01	+0.01	—	73
Avg firm size (local ccy)	+0.49	−0.77	−0.17	72
Avg firm size (USD)	+0.15	−0.12	−0.47	72

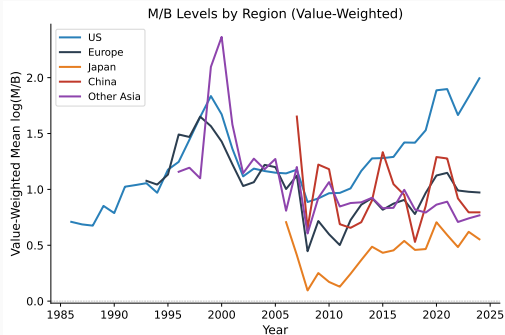
Savings/GNI is the **only** significant predictor of P/E ($t = 3.01$). Institutional variables are null. Local-currency size was a currency artifact. Savings/GNI is orthogonal to ROA, so it does not absorb the ROA–P/E puzzle.

Regional Patterns

Regional P/E and M/B Over Time



Value-weighted P/E



Value-weighted M/B

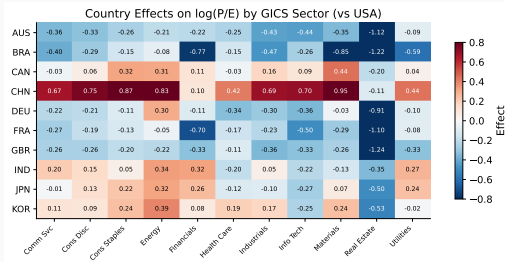
US dominates in VW (tech firms). Japan's M/B discount is stark (book value conservatism), but P/E is near Europe. China P/E volatile and declining since 2015.

The US Valuation Premium (Recent 5 Years)

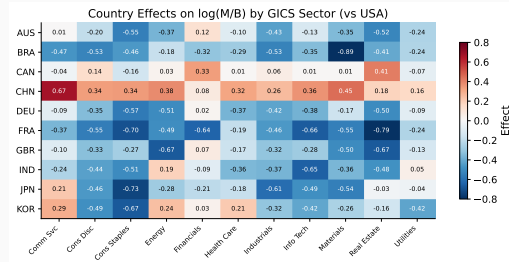
US minus	P/E gap		M/B gap	
	EW	VW	EW	VW
Europe	+18%	+24%	+33%	+123%
Japan	+36%	+54%	+98%	+258%
China	-48%	+7%	-18%	+132%
Other Asia	+7%	+48%	+50%	+190%

- VW gaps are much larger than EW — US premium is concentrated in large firms (tech)
- Japan: P/E gap +54% but M/B gap +258% — most of the M/B gap is book equity measurement
- China: EW gap is -48% (China P/E higher, driven by small firms); VW gap only +7%

Industry × Country Interaction



P/E country effects by sector



M/B country effects by sector

Country effects vs USA by GICS sector. Real Estate shows large negative effects across most countries. China premium is broad-based.

Summary and Next Steps

Stylized Facts

1. **Country dominates valuation** — 65% of R^2 for P/E (Shapley). Extends Heston–Rouwenhorst (1994) from returns to levels.
2. **P/E is cleaner than M/B** — M/B contaminated by leverage (+0.90***) and accounting (Japan: M/B -0.35 vs P/E $\approx$ 0).
3. **Valuation dispersion is diverging** — P/E dispersion $\times 2.7$ despite financial globalization.
4. **Negative ROA–valuation correlation across countries** ($\rho \approx -0.4$). Within countries, the sign flips positive (+0.80). A Simpson's paradox driven by size conditioning.
5. **National savings rate predicts P/E** ($r = +0.34$, $t = 3.01$). The only significant macro predictor. More savings $\Rightarrow$ more capital chasing equities $\Rightarrow$ higher multiples.
6. **Standard institutional variables are null** — La Porta, financial development, GDP per capita: all near zero.

Research Questions

1. **Why is P/E dispersion diverging?** ($\times 2.7$) — US tech? Post-GFC monetary policy? Capital flows?
2. **Can the savings channel be formalized?** — Savings/GNI predicts P/E but is orthogonal to ROA. Can we build a model where savings rates affect discount rates and therefore valuations?
3. **Do low-valuation countries earn higher returns?** — Bekaert–Harvey, Fama–French international. If so, valuation gaps reflect risk premia.
4. **What explains China's P/E premium?** — Highest P/E despite moderate M/B. Retail investors? Capital controls? Savings rate (46% of GNI)?
5. **EV/EBITDA as third metric** — avoids both the book equity issues (M/B) and the profitable-firms restriction (P/E).

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q. Our results: weak for P/E.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.

Savings and asset prices:

- Abel (2003, *Brookings*) — effect of baby boomers' saving on asset prices.
- Caballero et al. (2008, *JME*) — global imbalances and asset price inflation.

Appendix

Regression Specifications

P/E (profitable firms, $NI > 0$):

$$\log(P/E)_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \beta_3 \Delta E_{t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

M/B (all firms):

$$\log(M/B)_{i,t} = \beta_1 \text{ROA} + \beta_2 \text{Lev} + \beta_3 \log(\text{Assets}) + \beta_4 \Delta E_{t-1} + \text{FE} + \varepsilon_{i,t}$$

ROA:

$$\text{ROA}_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

$\text{FE} = \gamma_c$ (country) + δ_j (GICS industry) + τ_t (year). SE clustered at country level. Spec 2: $\gamma_{c,t}$ (country×year) replaces $\gamma_c + \tau_t$.

P/E and M/B Regression Results

	P/E (profitable firms)	
	Spec 1	Spec 2
Leverage	−0.017 (0.102)	−0.013 (0.100)
log(A_t)	−0.076** (0.030)	−0.078** (0.031)
Lag ΔE	−0.016*** (0.003)	−0.016*** (0.003)
N / R^2	428K / 0.142	428K / 0.163

	M/B (all firms)	
	Spec 1	Spec 2
ROA	0.259 (0.158)	0.225 (0.154)
Leverage	0.901*** (0.060)	0.902*** (0.063)
log(A_t)	−0.074*** (0.027)	−0.077*** (0.028)
N / R^2	593K / 0.219	593K / 0.246

Spec 1: Country + Industry + Year FE. Spec 2: Country×Year + Industry FE. SE clustered by country.

Market Cap Construction (Detail)

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: $\text{Market Cap} = \text{cshoi}$ (shares outstanding) $\times$ `prccm` (month-end price)
- **Pitfall:** firms have multiple securities across exchanges and currencies

Example — SAP SE (`gvkey=103487`) in Dec 2022:

`iid=02W` `EUR` 96.39 ← correct

`iid=05W` `HUF` 38,620 ← wrong currency!

`iid=06W` `MXN` 2,009 ← wrong currency!

- **Solution:** use `prirow` field (primary security designation) to select the correct `iid`

All data in local currency (millions). Cross-country *levels* absorbed by country FE.

Accounting Standards

Compustat Global reports accounting standard via acctstd:

Standard	Share
IFRS	59.7%
Domestic GAAP	39.3%
US GAAP	0.2%
Other / missing	0.8%

US firms: all US GAAP.

Country variation:

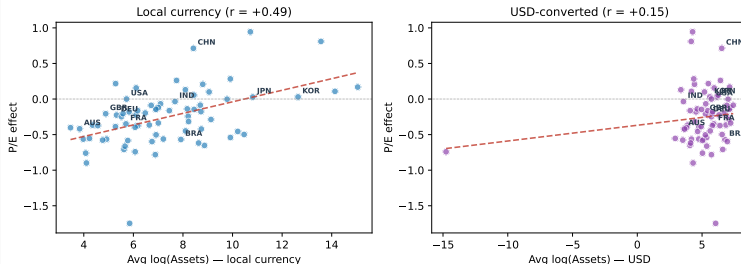
- EU: mandatory IFRS since 2005
- Japan: 93% domestic GAAP
- China: 96% domestic GAAP
- Korea: 99% IFRS (since 2011)

Book equity (ceq) particularly affected: IFRS vs. local GAAP differ on intangibles and fair value.

Country FE absorb time-invariant accounting differences. Concern: within-country switches (EU 2005).

Firm Size: Local Currency vs USD

Does Currency Conversion Change the Size-P/E Relationship?



	Size vs P/E	Size vs ROA	Partial $r(\text{ROA}, \text{P/E} \mid \text{size})$
Local currency	+0.49	-0.77	-0.17
USD-converted	+0.15	-0.12	-0.47

Local currency: JPN $\log(at) = 10.8$, KOR 12.6, IDN 13.6. In USD: 6.1, 5.6, 4.2. After conversion, size no longer absorbs the ROA-P/E link.